

FRANKLINE MISANGO OYOLO

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Education

The University of Hong Kong

Aug 2022 – Jun 2027

BEng. Artificial Intelligence & Data Engineering — GPA: 3.73/4.3

Hong Kong SAR

Professional Experience

The Swiss Federal Institute of Technology - Lausanne

Jul 2025 – August 2025

PhD Research Assistant - Machine Learning and Advanced Analytics

Switzerland

- Developed Diffusion Convolutional RNN Network for Swiss bike-share forecasting using EPFL HPC clusters, achieving 27% RMSE
- Engineered multi-scale OSM feature integration (1500m radii) across 1k+ stations for spatial dependency modeling
- Designed attention-based interpretability framework revealing station-type specific patterns (local vs regional focus)

BASF SE

Jan 2025 – Jun 2025

Business Intelligence Analyst - Global Supply Chain and Digital Business Solutions

Hong Kong

- Architected real-time supply chain pipelines using Alteryx, KNIME and SAP, processing 14TB+ logistics data while reducing stockouts through automated replenishment signals
- Engineered hybrid search system using Azure Cognitive Search, reducing NLP query latency 42% for 5k+ customer support officers
- Visualized supply chain KPIs with Power BI, enhancing executive decision-making through interactive dashboards tracking 120+ metrics across 30 countries

Manulife Financial Corporation

Jun 2024 – Aug 2024

Data Governance Summer Analyst

Hong Kong

- Engineered hybrid contextual generation pipeline using LangChain + Azure OpenAI GPT-4 Turbo, achieving 92.4% description accuracy across 312k Purview assets (38% error reduction vs manual)
- Implemented dynamic few-shot learning with Azure Cognitive Search index (1.2M reference docs) to cut manual metadata remediation by 650h/month
- Developed automated GPT-Lint test suite with PyTest + Azure ML, increasing compliance with 23 Manulife global standards (GDPR/CCPA) by 41% YoY

Department of Architecture, University of Hong Kong

Dec 2023 – Apr 2024

PhD Research Assistant - Machine Learning and Analytics

Hong Kong SAR

- Engineered Meta SAM transformer-based instance segmentation pipeline (ViT-H backbone, 632M parameters) achieving 91.8% IoU on 1000+ architectural floor plans for aeration feature analysis in microbial containment studies
- Fine-tuned DeepLab 3+ with Xception-65 backbone (26.5M parameters) on segmented thermal imagery dataset (n=4,800) for temporal prediction of ventilation efficiency, achieving 0.87 R² on 3-year forecasts
- Architected Kubernetes-orchestrated MLOps pipeline with COCO annotation standard, reducing data preprocessing time by 78% and enabling automated multi-GPU training across 3 distributed DL workloads

TalentLabs Asia Limited

Jun 2023 – Aug 2023

Data Engineer

Hong Kong SAR

- Built Analytics dashboard using LookerML processing 12k+ user events, improving UI conversion by 75% through A/B testing
- Automated GCP ETL pipelines with Prefect orchestrator, reducing weekly reporting time from 20h to 2h
- Developed distributed web scraper harvesting 8k+/day HK job postings via Python Scrapy/Selenium with 2CAPTCHA integration

Projects

FeenQR | C#, AI (RAG), Data Analysis

2025

- Developed RAG-based system using Semantic Kernel for quantitative research, analyzing YouTube videos and market trends to generate trading signals
- Implemented AI-driven data analysis and signal generation, relevant to materials macro modeling and strategic projects
- Utilized C# for scalable AI pipelines, tying into Python/SQL skills and Power BI/DAX for business insights

Algorithmic Trading and HFT Research | Python, Jupyter, Mathematical Modeling

2025

- Researched quantitative problems including market making and pricing models using Jupyter Notebooks
- Demonstrated statistical modeling, time series analysis, and optimization for supply chain analytics and cost transparency
- Applied Python for mathematical modeling, matching resume's analytics and ML expertise

Trading-Terminal | Python, AWS, Data Processing

2024

- Developed quantitative pre-trading research tool for stocks, crypto, bonds, and ETFs, hosted on AWS
- Implemented data analysis and research automation, including ETL and querying financial data
- Applied cloud-based BI techniques, aligning with supply chain analytics and forecasting needs

Technical Skills

Business Tools: Alteryx, KNIME, SAP, Microsoft Power Apps, Databricks, Azure Data Factory

Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional

Analytics: Python (Pandas/NumPy/Scikit-learn), R, SQL, MATLAB, Statistical Modeling, Time Series Analysis

ML/AI: TensorFlow, PyTorch, XGBoost, CuDA, Deep Learning, Natural language processing

Visualization: Tableau, Power BI, Matplotlib/Seaborn, D3.js, Plotly, LookerML, RapidsML

Cloud/Data: AWS, Azure, Snowflake, PostgreSQL, Apache Spark, Kafka, ETL Pipelines